

Homework 1

Uriel Benítez Linear Algebra Sep 17, 2026

Problem 1

(a)

Final answer: 24

(b)

$$(1/2) w = [-2, 1, 3]^T$$

$$v + (1/2) w = [1, 4, -2]^T + [-2, 1, 3]^T = [-1, 5, 1]^T$$

$$||v + (1/2) w|| = (-1)^2 + (5)^2 + (1)^2 = 1 + 25 + 1 = 27$$

[ANSWER: 27]

(c)

$$\text{use } \cos \theta = (u \cdot w) / (||u|| ||w||).$$

$$u \cdot w = (2)(-4) + (-1)(2) + (3)(6) = -8 - 2 + 18 = 8$$

$$||u|| = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{4 + 1 + 9} = \sqrt{14}$$

$$||w|| = \sqrt{(-4)^2 + 2^2 + 6^2} = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

$$\cos \theta = 8 / (\sqrt{14})(2\sqrt{14}) = 8 / 28 = 2/7$$

$$\theta = \arccos(2/7) = 73.3985\dots^\circ$$

[ANSWER: $\theta \approx 73.40^\circ$]

Problem 2

(didn't get to this)

Problem 3

(a)

Augmented matrix:

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 3 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_1: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 3 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$R_2 \rightarrow -R_2: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_2: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 10 & | & 10 \end{bmatrix}$$

$$R_3 \rightarrow (1/10)R_3: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & | & 1 \end{bmatrix}$$

Back substitute: $x_3 = 1$, then $x_2 = -4 + 3(1) = -1$, then $x_1 = -1 - 2(-1) + 1 = 2$.

One solution only.

[ANSWER: $x = [2, -1, 1]^T$]

(b)

$$\begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 5 & 1 & | & 6 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1: \begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_1: \begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2: \begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 2R_2: \begin{bmatrix} 1 & 0 & -7 & | & 8 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Column 3 has no pivot, so z is free and there are infinitely many solutions.

[ANSWER: infinitely many solutions, z is free]

Problem 4

(a)

Let x_1 = mass of Alloy 1 used, in kg.

Let x_2 = mass of Alloy 2 used, in kg.

Let x_3 = mass of Alloy 3 used, in kg.

The 100 kg of target alloy holds 36 kg copper, 44 kg zinc, and 20 kg nickel, and metal in = metal out, so:

$$\text{copper: } 0.5x_1 + 0.2x_2 + 0.4x_3 = 36$$

$$\text{zinc: } 0.3x_1 + 0.7x_2 + 0.2x_3 = 44$$

$$\text{nickel: } 0.2x_1 + 0.1x_2 + 0.4x_3 = 20$$

(c)

Times 10 to kill the decimals:

$$5x_1 + 2x_2 + 4x_3 = 360$$

$$3x_1 + 7x_2 + 2x_3 = 440$$

$$2x_1 + x_2 + 4x_3 = 200$$

$$R_1 \rightarrow R_1 - 2R_3: \begin{bmatrix} 1 & 0 & -4 & | & -40 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 2 & | & 440 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & 4 & | & 200 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1: \begin{bmatrix} 1 & 0 & -4 & | & -40 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 7 & 14 & | & 520 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & 4 & | & 200 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_1: \begin{bmatrix} 1 & 0 & -4 & | & -40 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 7 & 14 & | & 520 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 12 & | & 280 \end{bmatrix}$$

$$R_2 \rightarrow (1/7)R_2: \begin{bmatrix} 1 & 0 & -4 & | & -40 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 2 & | & 520/7 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 12 & | & 280 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2: \begin{bmatrix} 1 & 0 & -4 & | & -40 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 2 & | & 520/7 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 10 & | & 1440/7 \end{bmatrix}$$

$$R_3 \rightarrow (1/10)R_3: \begin{bmatrix} 1 & 0 & -4 & | & -40 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 2 & | & 520/7 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & | & 144/7 \end{bmatrix}$$

$$x_3 = 144/7 \approx 20.57$$

$$x_2 = 520/7 - 2(144/7) = 232/7 \approx 33.14$$

$$x_1 = -40 + 4(144/7) = -40 + 576/7 = 296/7 \approx 42.29$$

[ANSWER: 42.29 kg of Alloy 1, 33.14 kg of Alloy 2, 20.57 kg of Alloy 3]

Problem 5

(a)

$$R_2 \rightarrow R_2 - 2R_1: \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 6 & 4 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_1: \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2: \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - R_2: \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

[ANSWER: RREF(A) = [120;001;000]]

(b)

$$\text{rank}(A) = 2$$

(c)

0 solutions: possible.

1 solution: impossible.

Infinitely many: possible.

Problem 6

(a)

Put v_1, v_2, v_3 in as columns and row reduce:

$$\begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_1: \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2: \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Two pivots but three vectors, so the set is linearly dependent.

The relation is $v_3 = v_2 - 2v_1$, since $[2, 1, 3]^T - [2, 0, 4]^T = [0, 1, -1]^T$.

v_1 and v_2 are not multiples of each other, so they span a plane through the origin.

[ANSWER: dependent; span is a plane in $\mathbb{R}^3$]

(b)

I'll use the determinant this time. Columns v_1, v_2, v_3 :

$$\begin{vmatrix} 1 & 0 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 0 & 1 & 1 \end{vmatrix}$$

Expand across the top row:

$$\det = 1(1 \cdot 1 - 0 \cdot 1) - 0(1 \cdot 1 - 0 \cdot 0) + 1(0 \cdot 1 - 1 \cdot 1)$$

$$= 1(1) - 0 + 1(-1)$$

$$= 1 - 1 = 0$$

The determinant is 0, so the columns are linearly dependent.

Since they are dependent but not all multiples of one vector, the span is a plane.

[ANSWER: dependent; span is a plane in $\mathbb{R}^3$]